

Dr. Varsha Singh

AI Researcher | Computer Vision | Generative AI | Human-Centered Computing

DST WISE-SCOPE Postdoctoral Fellow

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Professional Profile

AI researcher with over 13 years of teaching experience and expertise in Artificial Intelligence, Computer Vision, Deep Learning, Generative AI, Multimodal Learning, and Human-Centered Computing. Research focuses on intelligent systems for healthcare, accessibility, education, and assistive technologies.

Research Interests

Artificial Intelligence • Machine Learning • Deep Learning • Computer Vision • Generative AI
• Multimodal Learning • Medical Image Analysis • Human-Centered AI • Assistive Technologies
• Image Captioning • Large Language Models • Cognitive Computing

Education

Ph.D. (Information Technology), IIIT Allahabad, 2024
Thesis: Attention Based Models for Generating Textual Description from Images and Visual Content from Text

M.Tech. (Computer Science & Engineering), UPTU (AKTU), 2011

B.Tech. (Computer Science & Engineering), UPTU (AKTU), 2005

Professional Experience

DST WISE-SCOPE Postdoctoral Fellow, IIIT Allahabad (2025–Present)

Temporary Faculty, MNNIT Allahabad (2024–2025)

Assistant Professor, JSS Academy of Technical Education (2007–2018)

Lecturer, NIET (2006–2007)

Selected Publications

- MAT2I: Enhancing Perceptual Authenticity in Text-to-Image Synthesis Using Multi-Attribute GANs (JAIR, 2025)
- Scene Description with Context Information Using Dense-LSTM (2023)
- Visual Content Generation from Textual Description Using Improved Adversarial Networks (MTAP, 2023)
- Indian Sign Language Recognition Using SURF with SVM and CNN (Array, 2022)

Current Projects

- DST WISE-SCOPE Project on AI-assisted communication and cognitive development for children with hearing impairment.
- Deep Learning for Facial Emotion Recognition.
- Attention-based Image Captioning.

Awards & Achievements

- DST WISE-SCOPE Fellowship
- MHRD Research Fellowship
- SERB-DST Travel Grant
- Best Paper Award (FICTA 2023)
- UGC-NET Qualified
- GATE (CSE) Qualified
- IEEE Reviewer
- Session Chair & Session Organizer (IHCI)
- Distinguished Academic Speaker

Teaching Areas

Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures, Operating Systems, DBMS, Python Programming

Technical Skills

Python, C/C++, MATLAB, PyTorch, TensorFlow, Keras, OpenCV, LaTeX, GitHub, Google Colab

Contact

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